

WOLLONGONG, Australia

WMO: 957480

Lat: 34.5639S

Lon: 150.7900E

Elev: 28

StdP: 14.68

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.9	39.3	29.8	23.9	57.1	31.6	26.0	55.4	33.6	55.6	30.3	57.0	2.4	260	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	88.3	68.4	83.7	69.2	81.0	69.7	74.6	81.6	73.2	79.8	72.0	78.2	15.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	120.2	78.1	71.0	114.5	77.0	69.9	109.9	76.1	38.1	81.6	36.9	80.1	35.8	78.3	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.2	24.4	20.8	DB	32.4	100.4	1.8	5.1	31.1	104.1	30.1	107.1	29.1	109.9	27.8	113.6	(4)
(5)			WB	31.7	76.8	1.7	1.4	30.5	77.8	29.5	78.6	28.5	79.4	27.3	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	70.9	70.4	68.1	63.5	58.3	54.7	53.2	54.3	57.9	61.5	64.9	67.8
	DBStd	7.36	4.44	3.91	3.70	3.79	3.52	3.34	3.61	3.70	4.52	4.86	4.65	4.46
	HDD50	16	0	0	0	0	0	3	9	4	0	0	0	0
	HDD65	1731	3	2	12	70	209	309	365	332	220	132	57	20
	CDD50	4426	649	570	562	405	258	144	109	138	236	357	447	551
	CDD65	668	187	152	110	25	2	0	0	1	6	25	54	106
	CDH74	4218	1198	818	503	160	22	0	3	9	125	290	408	682
	CDH80	962	285	148	71	26	0	0	0	0	29	95	129	179
Wind	WSAvg	8.6	8.5	8.0	7.7	7.1	8.3	8.6	9.4	9.8	9.2	8.9	8.6	8.7
Precipitation	PrecAvg	44.3	4.0	4.8	4.8	4.3	3.5	4.0	1.8	3.1	2.4	3.2	4.5	3.1
	PrecMax	73.9	18.3	19.2	18.8	17.6	15.5	29.4	7.6	25.9	10.3	12.4	30.5	11.5
	PrecMin	18.9	0.8	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.1	0.6	0.2
	PrecStd	14.0	3.2	4.4	3.9	4.5	3.5	5.0	1.8	4.9	2.5	3.0	5.0	3.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.0	91.7	87.0	83.6	76.7	70.3	72.2	74.9	83.9	90.2	91.7	92.4
		MCWB	73.6	75.0	71.2	63.3	60.7	57.0	56.0	56.0	60.6	63.2	66.8	65.6
	2%	DB	85.8	83.3	81.0	77.7	72.8	66.8	67.4	70.1	77.4	81.5	82.8	84.6
		MCWB	72.0	72.4	70.7	65.7	60.7	56.8	54.1	55.5	58.4	62.8	66.7	67.5
	5%	DB	82.1	80.6	78.7	75.2	70.1	64.9	64.6	66.8	72.4	76.0	77.9	80.1
		MCWB	72.6	71.6	69.8	65.1	59.6	55.7	53.0	54.5	57.8	62.6	67.7	69.1
	10%	DB	79.8	78.7	76.6	72.9	67.8	62.8	62.2	64.2	68.5	72.5	74.7	77.2
		MCWB	71.3	70.5	68.7	64.2	58.3	54.6	52.3	53.5	57.7	62.6	66.0	68.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	76.6	74.4	70.7	65.4	61.1	60.4	60.8	64.3	69.1	72.6	73.9
		MCDB	84.9	86.6	79.6	77.5	70.6	65.3	66.2	67.1	74.1	75.9	81.3	81.6
	2%	WB	74.8	74.4	72.7	68.3	63.1	59.1	57.3	58.5	62.5	67.2	70.5	72.1
		MCDB	82.3	80.9	78.5	74.2	69.0	63.3	62.5	65.2	70.6	74.4	78.2	79.2
	5%	WB	73.5	72.9	71.2	66.9	61.2	57.7	55.4	56.8	60.9	65.5	68.7	70.7
		MCDB	80.6	78.6	77.0	72.7	67.8	62.2	61.1	63.9	68.7	73.1	75.6	77.4
	10%	WB	72.1	71.6	69.8	65.3	59.5	56.4	53.7	55.0	58.9	63.7	67.1	69.3
		MCDB	78.6	76.9	75.5	70.9	66.1	61.0	60.3	61.9	66.5	71.1	73.1	75.5
Mean Daily Temperature Range	5% DB	MDBR	15.5	14.4	15.4	18.0	19.0	17.0	19.0	20.0	20.5	19.1	16.8	16.4
		MCDBR	21.3	18.9	19.5	22.6	23.4	21.0	23.4	25.8	29.2	28.5	25.3	24.0
		MCWBR	10.4	9.7	11.2	12.3	13.1	12.2	12.7	14.0	14.2	14.2	12.4	11.0
	5% WB	MCDBR	18.0	16.3	17.3	19.8	21.6	17.1	20.2	22.5	23.9	22.3	20.4	19.0
MCWBR		9.7	8.9	10.3	11.9	13.4	11.1	12.7	14.0	14.3	13.1	11.7	10.9	
Clear-Sky Solar Irradiance	taub	0.400	0.389	0.365	0.336	0.318	0.308	0.298	0.307	0.339	0.357	0.388	0.402	
	taud	2.430	2.487	2.528	2.584	2.584	2.610	2.622	2.589	2.500	2.482	2.430	2.400	
	Ebn at Noon	297	294	289	281	269	264	274	287	293	300	299	298	
	Edh at Noon	39	35	32	27	23	21	22	26	32	35	38	40	
All-Sky Solar Radiation	RadAvg	2055	1741	1489	1202	944	737	866	1148	1544	1813	1928	2104	
	RadStd	156	154	125	102	57	68	69	64	93	149	169	167	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	+0.77	N/A	N/A	+1.35	N/A	N/A	-2	-153	+328	+167

Nomenclature: See separate page